

Jeehong Kim

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RESEARCH INTEREST

I am interested in designing broadly applicable machine learning techniques for graph-based problems.

- **Keywords:** Graph Representation Learning, Geometric Deep Learning

EDUCATION

Seoul National University, Seoul, South Korea

- M.S.-Ph.D. Integrated Program in Data Science Mar 2023 – Present
 - Research Interest: Graph Representation Learning
 - Advisor: Prof. Hyunwoo Park
- Planning to fulfill military service as a *Technical Research Personnel* starting around September 2027

Korea University, Seoul, South Korea

- B.S. in Industrial Management Engineering Mar 2019 – Feb 2023
 - GPA: 3.76 / 4.50

RESEARCH EXPERIENCE

Materials Intelligence Lab, LG AI Research

- Research Intern Feb 2026 – Present
 - Research Interest: Molecule Property Prediction, Molecular Foundation Model, Geometric Deep Learning
 - Advisor: Dae-Woong Jeong, Sumin Lee, Sung Moon Ko

DSAIL (Data Science & Artificial Intelligence), KAIST

- Undergraduate Research Intern Jun 2022 – Aug 2022
 - Research Interest: Graph Neural Network, Recommendation System
 - Advisor: Prof. Chanyoung Park

AIDA Lab (Artificial Intelligence & Data Analytics), Korea University

- Undergraduate Research Intern May 2021 – Jan 2022
 - Research Interest: Anomaly Detection, GAN
 - Advisor: Prof. Sungwon Han

PUBLICATIONS

(†: Equal contribution)

CONFERENCE

- [C1] Redefining Maritime Anomaly Detection via Equation-Grounded Synthetic Anomalies
Youngseok Hwang[†], Sungho Bae[†], Dohun Lee, Jaeun Seo, **Jeehong Kim**, Wonhee Lee, Hyunwoo Park
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026 AI for Science Track (Oral)**)

JOURNALS

- [J2] STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection
Jeehong Kim[†], Minchan Kim[†], Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, Hyunwoo Park
IEEE Access (2025.09)
- [J1] FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels
Yooyeun Kim[†], **Jeehong Kim**[†], Wonhee Lee, Hyunwoo Park, Deuk Jae Cho
PLoS One (2025.05)

WORKSHOPS

- [W2] Spatio-Temporal Graphs Beyond Grids: Benchmark for Maritime Anomaly Detection
Jeehong Kim[†], Youngseok Hwang[†], Minchan Kim, Sungho Bae, Hyunwoo Park
NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery
- [W1] Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies
Jeehong Kim[†], Minchan Kim[†], Jaeseong Ju, Youngseok Hwang, Wonhee Lee, Hyunwoo Park
AAAI 2025 Workshop in Anomaly Detection in Scientific Domains

PROJECTS

Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Vessel Behavior Anomaly Detection Oct 2024 – Dec 2025
 - Anomaly detection using a GNN-based framework on a customized dataset
 - (Published) *Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies*
 - (Published) *STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection*

	- Visualization for Ship Collision Risk Assessment Mar 2024 – May 2025 - Development of a decision supportive interactive system for ship collision avoidance (Published) <i>FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels</i>	
AWARDS & SCHOLARSHIPS	Silver Reviewer, ICML 2026 - Recognized as silver reviewer (top 50% of reviewers) Best Poster Award Dec 2025 - NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery BK 21 Scholarship, Brain Korea Mar 2025 – Present SNU Medical AI Scholarship, Seoul National University Aug 2024 – Aug 2025 Veritas Scholarship, Korea University Feb 2022 - Research on detection of anomalies in the manufacturing process - Advisor: Prof. Sungwon Han HAICON2021, Korea Institute of Information Security & Cryptology (KIISC) Nov 2021 - Construction of an AI model for identifying time-series based anomalies in industrial control systems - Ranked 6th out of 177 teams Industrial Engineering Project Competition, Korean Institute of Industrial Engineers (KIIE) Oct 2021 - Design of a dynamic corridor network for K-UAM - Received participation award in industrial engineering project competition Data Visualization Competition, Korea University, Yonsei University, Intel Aug 2021 - Visualization of patient and their deductible premium relationship - Received grand prize	
PROFESSIONAL SERVICES	Conference Review - Association for the Advancement of Artificial Intelligence (AAAI) 2026 – Present - International Conference on Machine Learning (ICML) 2026 – Present - Neural Information Processing Systems (NeurIPS) 2026 – Present - Learning on Graphs Conference (LOG) 2025 – Present Workshop Review - AI for Science(AI4Science)@NeurIPS 2025 – Present Journal Review - IEEE Access 2025	
ACTIVITIES	Lab Captain, Data Intelligence and Learning Lab (DIAL) Dec 2024 – Jan 2026 Advisor: Prof. Hyunwoo Park Samsung Dream Class, Samsung Welfare Foundation Sep 2021 – Feb 2022 Korea-China Leadership Program, Korea Foundation for Advanced Studies (KFAS) Jul 2019	
TEACHING EXPERIENCE	Teaching Assistant - K-Digital training program at Seoul National University Jun 2024 - Python data analysis training program for Seoul National University faculty members Jan 2024	
TALKS AND SEMINARS	FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels 2024 INFORMS Annual Meeting (Contributed Session) Oct 2024	
REFERENCES	Prof. Hyunwoo Park, Associate Professor, Seoul National University Email: hyunwoopark@snu.ac.kr	

[CV compiled on 2026-10-07]